

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTETC503 - ANALOG COMMUNICATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 According to the Sampling Theorem, a signal can be completely represented if the sampling frequency is:
 - a) Equal to the highest frequency of the signal
 - b) Less than the highest frequency of the signal
 - c) At least twice the maximum frequency of the signal
 - d) Half the maximum frequency of the signal
- 2 The main advantage of SSB-SC over DSB-FC is:
 - a) Requires less bandwidth and power
 - b) Easier to demodulate
 - c) Better quality of transmission
 - d) High redundancy
- 3 VSB (Vestigial Sideband) modulation is mainly used in:
 - a) Telephone communication
 - b) AM broadcasting
 - c) Television transmission
 - d) Satellite communication
- 4 For 100% modulation in AM, the modulation index μ is:
 - a) 0.1
 - b) 0.5
 - c) 1
 - d) 2
- 5 Which of the following is true for FM compared to AM?
 - a) FM requires larger bandwidth
 - b) FM has better noise immunity
 - c) FM requires more complex circuitry
 - d) All of the above
- 6 The modulation index for FM is defined as:
 - a) $\Delta f / f_m$
 - b) $f_m / \Delta f$
 - c) f_c / f_m
 - d) P_c / P_s
- 7 In a superheterodyne receiver, the frequency produced by mixing the carrier and local oscillator is called:
 - a) Carrier frequency
 - b) Modulating frequency
 - c) Intermediate frequency (IF)
 - d) Beat frequency
- 8 Multiplexing allows:
 - a) Multiple signals to share the same channel
 - b) Conversion of digital signals into analog signals
 - c) Reduction of noise in transmission
 - d) Increase in antenna size
- 9 Thermal noise is also called:
 - a) Flicker noise
 - b) White noise
 - c) Shot noise
 - d) Burst noise

- 10 Automatic Gain Control (AGC) in receivers is used to:
- a) Increase receiver bandwidth
 - b) Keep output volume constant for varying signal strength
 - c) Improve image frequency rejection
 - d) Control oscillator frequency
- 11 Noise temperature is a measure of:
- a) Noise in terms of equivalent temperature
 - b) Thermal efficiency of a receiver
 - c) Heat produced by electronic devices
 - d) Bandwidth of noise spectrum
- 12 Signal-to-Noise ratio (SNR) is defined as:
- a) Ratio of noise power to signal power
 - b) Ratio of signal power to noise power
 - c) Difference between signal and noise
 - d) Product of signal and noise

Q2. Solve the following.

- A) Define communication, Draw and explain basic block diagram of communication system. 6
- B) Explain DSB SC variant of AM. Derive the expression for the total transmitted power of a DSB-SC signal. 6

Q3. Solve the following.

- A) Define the terms : (a) Pre-emphasis, (b) De-emphasis, (c) Narrowband FM and (d) Wideband FM 6
- B) Compare Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM). 6

Q4. Solve Any Two of the following.

- A) Define: Sensitivity, Selectivity, Fidelity, Image Frequency 6
- B) State and prove the Sampling Theorem. Give its significance in communication systems. 6
- C) Differentiate between simplex, half-duplex, and full-duplex communication systems with examples 6

Q5. Solve Any Two of the following.

- A) Define Noise. What are different sources of noise? Explain in detail. 6
- B) What is balanced modulator? Sketch a balanced modulator circuit & explain its working. Derive the expression for its output stage. 6
- C) In an FM system, when the Audio Frequency (AF) is 700 Hz, and the AF voltage is 1.2 V, the deviation is 2.4 kHz. If the AF voltage is now increased by 3.6 V, what is new deviation? If the AF voltage is further raised to 7.2 V while the AF is dropped to 500 Hz, What is the deviation? Find modulation index in each case. 6

Q6. Solve Any Two of the following.

- A) State & Explain the difference between simple AGC & Delayed AGC. 6
- B) Define Noise Figure, Noise Factor, and explain their relationship in evaluating the noise performance of electronic devices 6
- C) A VSB transmitter that transmits 20 % of the opposite sideband along with the desired sideband radiates 125 W when the modulation percentage is 70%. Calculate the total power required to transmit the same message using an AM transmitter. 6

*** End ***